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AI-Powered Autonomous Drilling Sets New Standards in a UAE Unconventional Project

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Abstract

Directional control and rig operations have traditionally relied on the expertise and focus of directional drillers and drillers, despite advancements in software and hardware solutions. Execution success remains highly dependent on human competency.

This paper presents the first successful deployment of an AI-powered autonomous directional drilling system integrated with a driller's advisory platform in an unconventional UAE field. The solution combines intelligent advisory systems with closed-loop steering automation to minimize human intervention in directional and on-bottom drilling workflows.

Implemented through collaboration between a service company and the field operator, the deployment aimed to redefine drilling performance and demonstrate scalability across multiple rigs. Key outcomes include improved drilling efficiency, trajectory accuracy, wellbore quality, and operational consistency—establishing a new benchmark for autonomous drilling in the region.

Introduction

Unconventional drilling operations in the UAE are characterized by high formation variability, tight directional tolerances, and the increasing need for efficiency and consistency in well delivery. The formation is highly interbedded and has a vast difference in ROP, formation tendency and BHA's directional work capability. The well trajectories have complex 3D profiles which require accurate directional control. These challenges place significant demands on both personnel and technology.

Conventional methods rely on manual processes driven by directional drillers (DD) and drillers, with DDs performing manual BHA tendency estimations and calculating steering control commands that are manually sent to the rotary steerable system (RSS) via pump speed changes. The drillers follow the drilling program manually, with no immediate indication of compliance. These human-dependent workflows introduce variability, limit scalability, and require constant supervision by experienced specialists.

As drilling activity intensifies and resource availability becomes more constrained, the industry is turning to automation and artificial intelligence (AI) to improve reliability, repeatability, and enable remote

execution. While digital tools and advisory systems have emerged in recent years, their field deployment remains limited, especially within unconventional assets.

This paper introduces a novel approach developed jointly by a service company and a UAE operator, detailing the motivations, enabling technologies, and integration strategy for reducing human intervention and improving well construction performance.

Field Context and Operational Environment

The UAE unconventional (UC) project was launched to unlock proven unconventional oil and gas resources and support national production targets. Key reservoirs include the Cenomanian carbonate Shilaif (UC) oil reservoir and the Upper Jurassic Diyab unconventional gas reservoir, comprising Jubaila, Hanifa, and Tuwaiq Mountain formations.

Development wells are drilled from multi-well pads (5–10 wells) using a three-section design after the conductor:

- 17.5/16 in. section – typically vertical
- 12.25 in. section – deviated for anti-collision
- 8.5 in. section – includes curve, landing, and horizontal intervals, requiring the most directional work and being the longest section

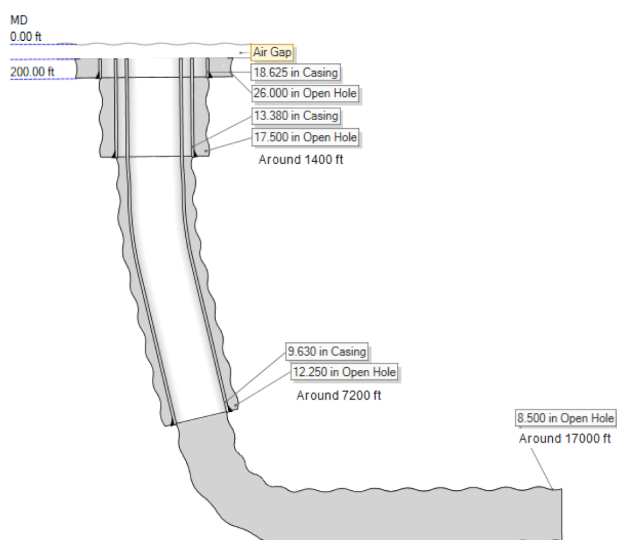


Figure 1—Typical wellbore geometry of an unconventional oil well.

The AI-powered autonomous directional drilling system, integrated with a driller's advisory platform, was primarily deployed in the 8.5 in. section. The project began in September 2024, during which well construction performance had already improved significantly prior to automation. Average ROP in the 8.5 in. section increased from 50–60 ft/hr to 80–100 ft/hr.

Autonomous Drilling System Overview

The deployed solution comprises two core components:

1. **Intelligent driller's advisory** - software applications and a human interface display providing real-time recommendations to the Driller on a set of optimum operational parameters to increase ROP while managing drilling dysfunctions and mitigating the impact of hard stringers. Intelligent driller's advisory solution comprises intelligent software and a driller's display.

2. **Autonomous directional drilling (ADD)** – software and hardware solutions enabling closed-loop execution of any directional trajectory with a Rotary Steerable System.

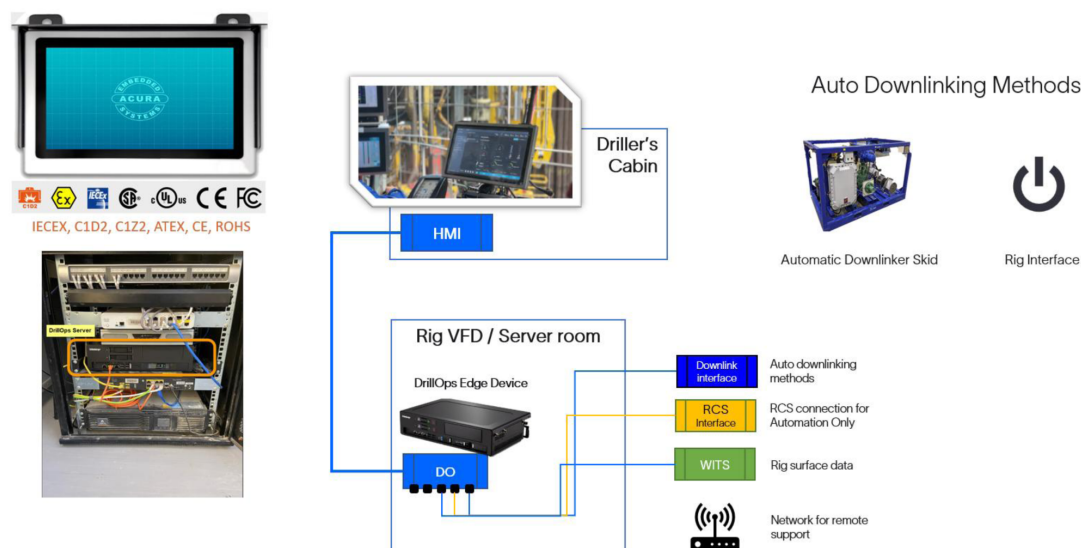


Figure 2—A layout of the integrated autonomous directional drilling and driller advisory solution.

Driller's advisory is based on an intelligent software module that delivers real-time recommendations to the driller to optimize drilling parameters like WOB and RPM to enhance on-bottom ROP, while continuously mitigating downhole risks. By eliminating the need for manual trial-and-error in setpoint selection, the system enhances operational efficiency and reduces non-productive time. The driller's advisory requires a depth-based drilling limits roadmap (WOB, RPM, ROP, Torque, etc.) ensuring that all recommendations remain within predefined safe operating windows. This functionality positions Driller's advisory as a compliance tool that enforces adherence to the drilling roadmap and facilitates early detection and correction of deviations from the agreed upon roadmap.

A key feature is real-time detection and mitigation of drilling dysfunctions, including stick-slip and lateral vibrations. Upon detection, the system recommends parameter adjustments to prevent tool failures and maintain drillstring integrity.

Autonomous directional drilling solutions enable autonomous execution of a directional trajectory to build wells in the most efficient and consistent manner possible, regardless of the rig, field, or trajectory.

Utilize AI-driven automation to:

-  → Execute your Plan
-  → Deliver the trajectory
-  → Optimize steering performance

-  AI Driven Directional Drilling steering
-  Automated tendency analysis and projections
-  Automated downlinking and motor steering
-  Detect and respond to steering dysfunction
-  Digitized DD operational model
-  Operate remotely with native cloud workflow
-  Automated steering performance insights

Figure 3—Key features of Autonomous Directional Drilling solution

The ADD system integrates three main elements:

1. Surface directional driller's advisory software.
2. Downlinking interface enabling a continuous feedback loop between the surface software application and a downhole rotary steerable system.
3. Downhole autonomous control system (embedded within the RSS)

Directional drilling advisor represents an integrated application that's able to execute an end-to-end directional drilling workflow, i.e. acquiring surface data from the rig and downhole data from RSS, performing BHA tendency analysis, and anticollision scanning to generate optimal steering recommendations according to user-defined constraints ultimately leading to drilling optimal wellbores.

As described in [1], the application is based on intelligent planning and intelligent execution to provide optimal recommendation to consistently execute the fastest, smoothest path with the lowest risk to achieve the directional drilling targets.

Downlinking interface is a minimal part of surface automation suite required to execute the directional drilling advisor's recommendations automatically thus minimizing disruptions to the drilling process and distraction of the driller and the directional driller. As depicted in Fig. 4., there are several methods available to downlink to RSS.

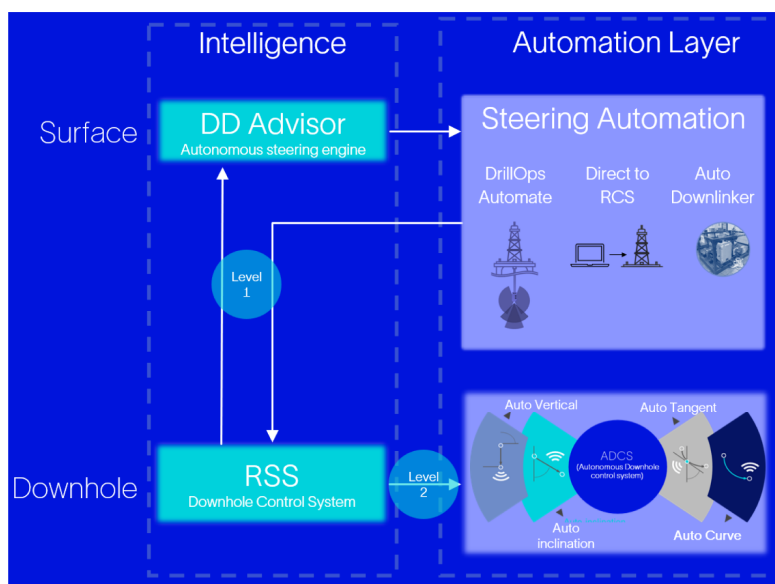


Figure 4—Key components of the Autonomous Directional Drilling technology

Since electronic interfaces for mud pumps or topdrive control were not available on the target rigs at the time of deployment, a flow bypass automatic downlinker was deployed to enable ADD.

Downhole autonomous control system is another method of directional drilling automation that relies solely on the sensors within RSS and its closed-loop control system enabling it to self-regulate its toolface and steering proportion to dynamically respond to the changes in drilling environment. As described in [2], there are several modes available:

- Hold inclination
- Hold inclination and azimuth
- Hold vertical
- Curve control

- Automatic kick off

The system supports native cloud connectivity for remote operations, allowing centralized technical teams to monitor and control activities from remote operations centers.

By combining surface and downhole automation, the solution delivers an end-to-end autonomous directional drilling workflow—from data acquisition and analysis to steering execution—ensuring optimal wellbore placement with minimal human intervention.

Deployment Methodology

Workflow Adaptation

Before deploying the Driller's Advisory and Autonomous Directional Drilling (ADD) solutions, a comprehensive review of the existing operating model was conducted to ensure seamless integration. Effective stakeholder engagement and change management were critical to breaking organizational silos and minimizing risk exposure.

Table 1—Simplified RACI chart with key activities and stakeholder responsibilities during deployment.

Tasks		Service Company - Field	Service Company - Office	Operator - Field/Office	Drilling Contractor
1	KPIs & Benchmarking				
	Define the KPIs & target offset wells for analysis	I	A/R	C	I
	Define the Drilling Roadmap (operating window)	I	C/I	A	I
	Develop the communication protocol	I	A/R	C	I
2	Equipment Installation & Commissioning				
	Edge server and Driller's HMI I&C	R	A/R	I	C/I
	Automatic Downlinker I&C	R	A/R	I	C/I
3	Pre-job Preparation				
	Attend the system awareness training	I	A	R	R
	Configure internet connection and validate data streaming settings	R	A	I	C/I
	Provide the definitive Drilling Roadmap	I	C/I	A	I
	Import & validate the BHA and the directional plan	R	A	I	I
	Configure DD Advisor application, set constraints and define drilling tunnel	R	A	I	I
	Configure Drilling Dysfunctions mitigation settings	R	A	I	I
4	Job Execution & Monitoring				
	Initial on-the-job training (hypercare)	R	A	I	R
	System utilization and adherence to recommendations	R/A	C	I	R
	Providing and communicating updates on drilling roadmap	I	R	R/A	I
	Hole cleaning monitoring (SPP, T&D) and response	I	I	A	R
5	Reporting & Job Evaluation				
	Prepare daily performance reports	R	A	I	I
	Prepare end of section & end of well reports	R	A	I	I
	Record and capture Stakeholder User feedback/LLs	R	A	I	I

A dedicated RACI (Responsible, Accountable, Consulted, Informed) matrix was developed to define roles and responsibilities across all stakeholders, including the operator, directional drilling service provider, and drilling contractor—both at the wellsite and in office environments.

Safety, Reliability, and Change Management

Transitioning from conventional to autonomous drilling requires a mindset shift and strong emphasis on training and communication. Risk assessments highlighted the need for intuitive system disengagement, allowing immediate manual intervention if required.

- The Driller's Advisory system does not control rig equipment, ensuring safe continuation of operations in case of malfunction.
- The ADD system elevates the directional driller's role to supervisory, where each current & upcoming steering decision is fully visible for monitoring. This allows the DD to focus on higher value tasks throughout their shift and only intervene with the system to train it in real time when personal experience may dictate unique, localized steering behavior patterns.

Management plays a critical role in setting expectations, communicating the value of system-based decision-making, and reinforcing the strategic importance of digital adoption for both operational performance and workforce development.

Field Deployment and Performance Results

Deployment Summary

The Driller's Advisory and Autonomous Directional Drilling (ADD) solutions were deployed across five wells targeting different sub-layers of the unconventional oil reservoir.

- Three wells utilized Driller's Advisory only.
- Two wells combined Driller's Advisory with ADD.

All wells achieved >70% utilization rates for the deployed technologies, demonstrating strong adoption and operational integration.

Impact on Rate of Penetration (ROP)

The deployment delivered significant improvements in drilling performance. Wells utilizing Driller's Advisory and ADD achieved record-breaking ROP values within their respective sub-layers, surpassing historical benchmarks.

Figures 5–9 illustrate ROP improvements across multiple sub-layers in the 8.5" hole section.

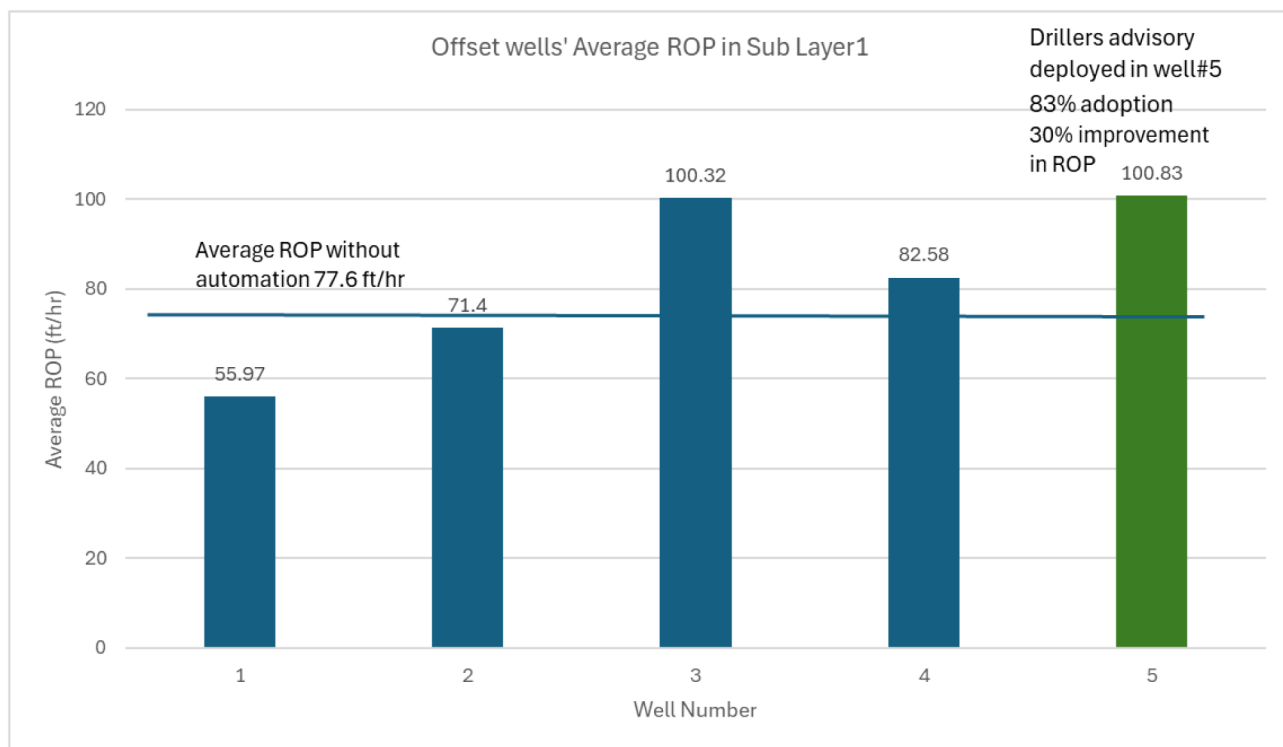


Figure 5—ROP improvement in Well 5 (8.5" in HS) in sub layer 1.

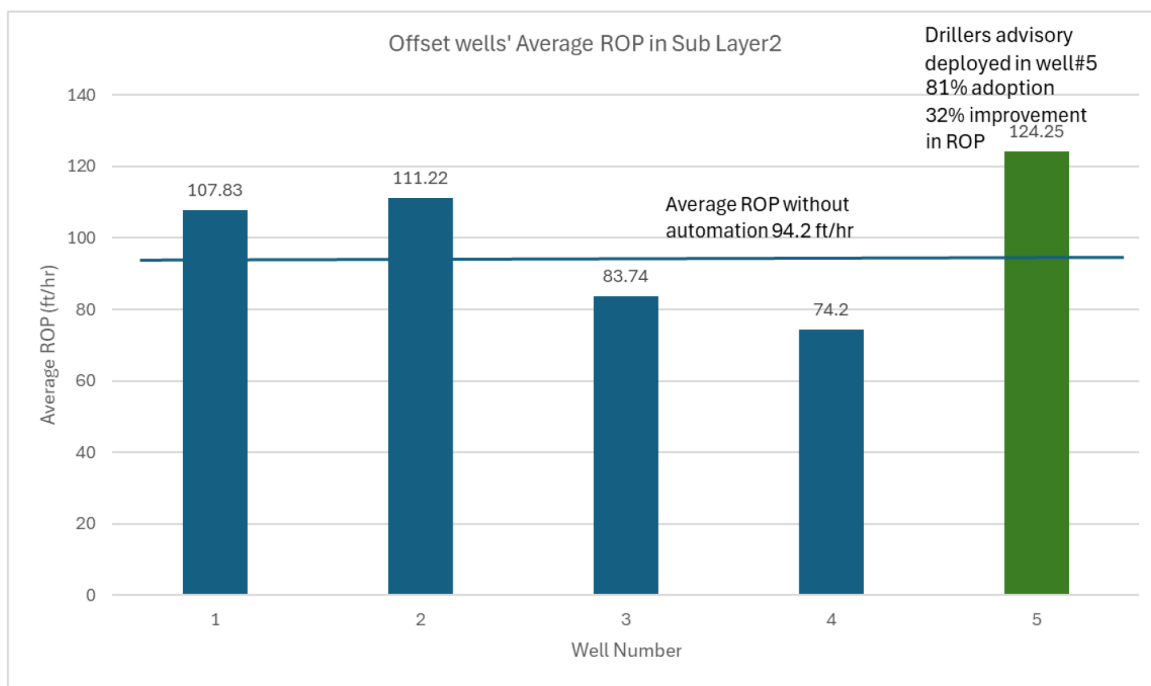


Figure 6—ROP improvement in Well 5 (8.5" in HS) in sub layer 2.

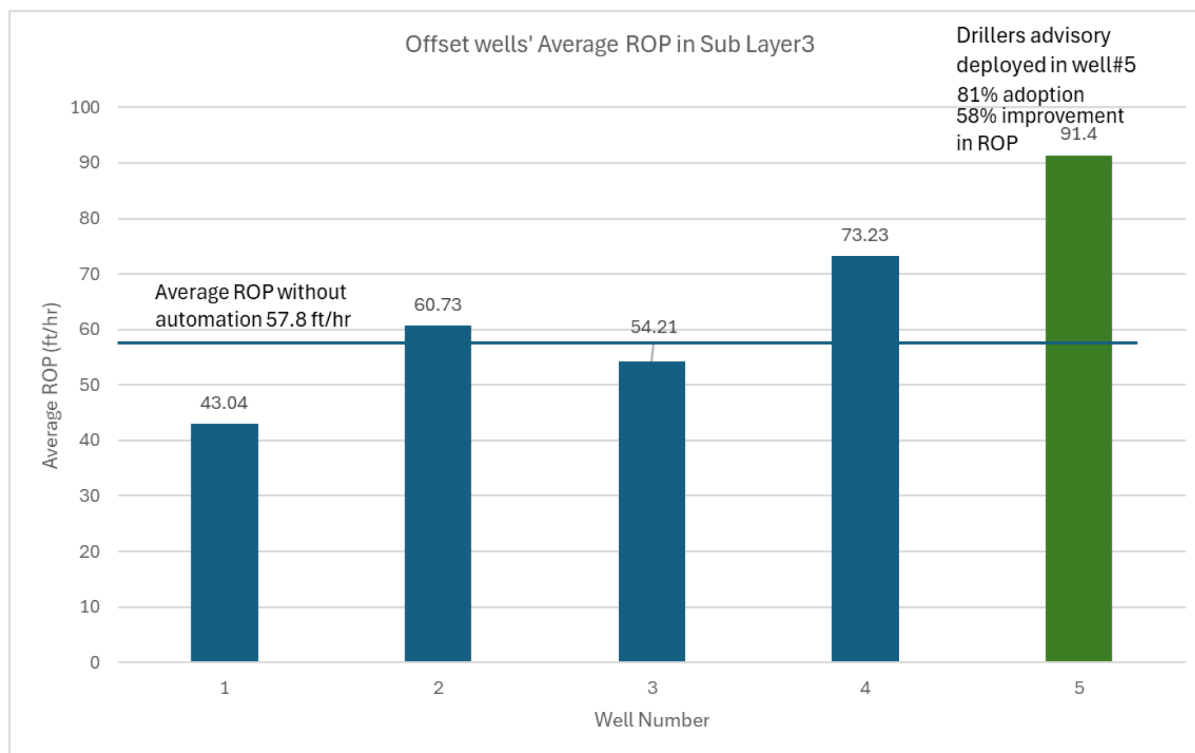


Figure 7—ROP improvement in Well 5 (8.5" in HS) in sub layer 3.

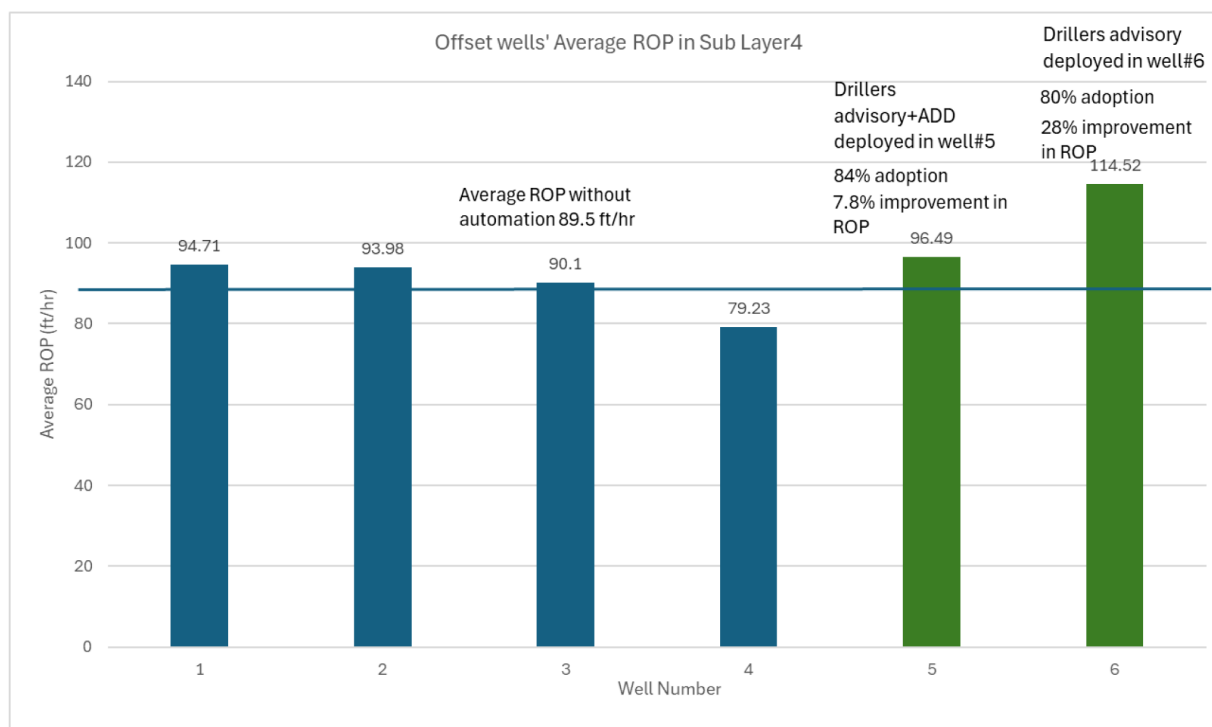


Figure 8—ROP improvement in Well 5 (8.5" in HS) in sub layer 4.

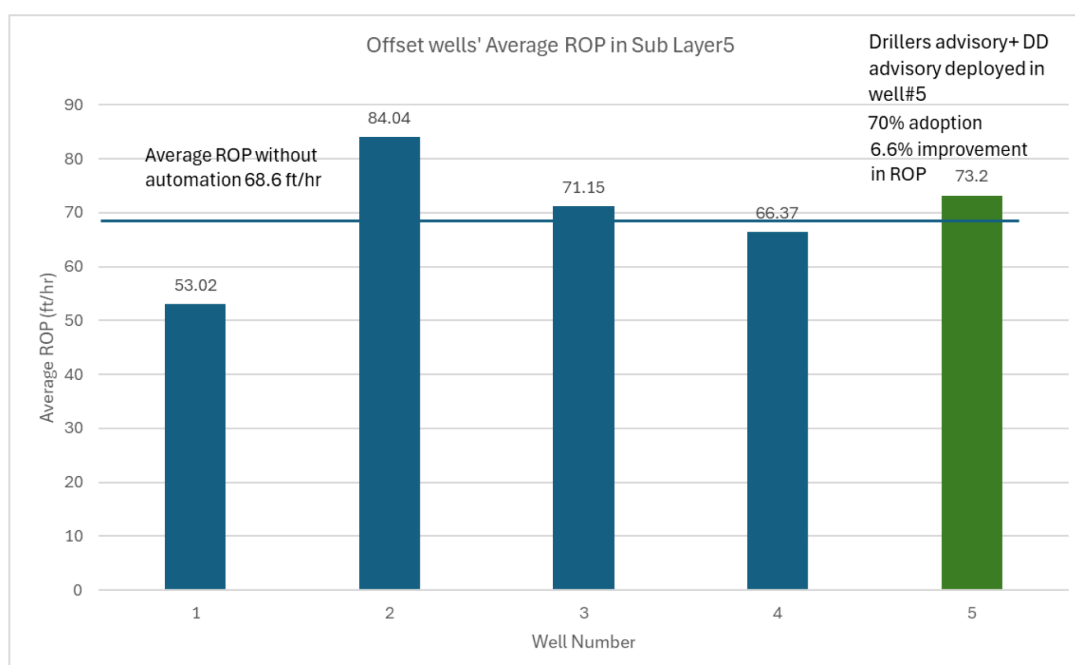


Figure 9—ROP improvement in Well 5 (8.5" in HS) in sub layer 5.

Reduction in Curve Tortuosity

The ADD system improved wellbore quality by minimizing deviations from the planned trajectory and reducing excess tortuosity through optimized steering decisions. Lower tortuosity facilitated smoother fracture string runs, reducing drag and associated buckling risks, and enabling the potential for longer lateral sections.

Figure 10 summarizes the reduction in average center-to-center (Ct-Ct) distance and excess tortuosity:

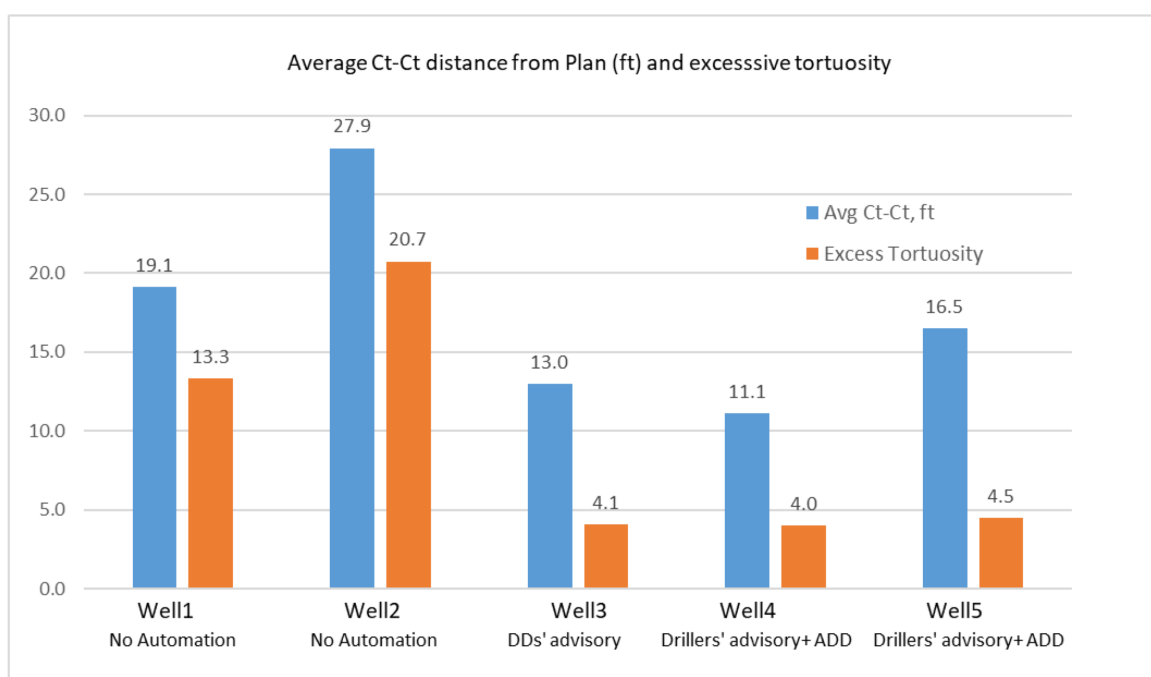


Figure 10—Reduction of Average Ct-Ct and Excess Tortuosity with ADD

Facilitated Fracture String Runs

Lower tortuosity further facilitated the fracture string running operations through the reduction of the excess drag with associated buckling risks, thus providing prospects of potential extension of the lateral section length, as shown in [table 2](#).

Table 2—Reduction of fracture string drag and improvement of fracture string buckling limit with ADD.

Offset Wells	Drag Increment vs Plan (%)	Buckling Depth Improvement (ft)
Well 1 (No automation)	5.0	-2000
Well 2 (No automation)	10.0	-250
Well 3 (Driller's Advisory)	1.2	+150
Well 4 (Advisory + ADD)	0.9	+500
Well 5 (Advisory + ADD)	1.5	+400

Downlink Time Observations

During this deployment, a mechanical flow-bypass automatic downlinker was used, configured with an 18-second bit period setpoint.

The number of downlinks required to execute the curve section of the trajectory ranged from 44 to 86, depending on formation response variations between wells and the configuration of DD Advisor drilling tunnel.

Since most automatic downlinks were executed while the bit was on-bottom, no correlation was observed between the number of downlinks and average ROP.

Consistency in Well Delivery

Higher adoption of Driller's Advisory and ADD correlates with reduced deviations from the optimal drilling performance curve. Scaling these technologies across multiple rigs will drive operational consistency, reduce variability caused by crew differences, and improve AFE predictability.

Reduction in On-Site Personnel

ADD shifts routine decision-making from the wellsite to remote operations, elevating the directional driller's role to supervisory. With reliable connectivity and basic BHA handling competency at the rig, it is feasible to remove directional drillers from the wellsite or significantly reduce their competency requirements. This deployment demonstrated that remote supervision is both practical and effective.

Enabling Remote Operations

Cloud-enabled remote operations centers play a critical role in scaling autonomous drilling solutions across the UAE unconventional program and beyond. This approach reduces HSE exposure, crew transportation, and associated carbon emissions, while improving operational oversight and efficiency [3].

Conclusions and Way Forward

This paper presented the successful deployment of a fully autonomous, closed-loop directional drilling system integrated with an AI-powered Driller's Advisory solution in UAE unconventional operations.

The approach is rig-agnostic, leveraging surface and downhole data to deliver actionable insights and autonomous execution capabilities.

A structured deployment methodology was outlined, emphasizing risk assessment, responsibility mapping, and stakeholder engagement as critical enablers for success.

Key findings are the following:

- Significant ROP Gains: When deployed with high compliance (>70%), the system delivered ROP improvements of 8–58% compared to baseline wells in the same sub-layers.
- Improved Wellbore Quality: Autonomous directional drilling reduced excess tortuosity, enabling smoother tripping operations and facilitating longer lateral sections through reduced drag and improved buckling limits.
- Operational Consistency: AI-driven automation minimized performance variability across crews, improving predictability and AFE accuracy.
- Remote Operations Enablement: The system allowed safe relocation of directional drillers to remote operations centers, reducing on-site personnel requirements and associated HSE exposure.
- Further gains in drilling efficiency are possible with the introduction of autonomous downhole control methods, primarily automated curve control. By enabling a closed-loop steering system embedded in RSS, it's possible to further reduce the excess tortuosity and improve performance through the reduction of downlink commands sent from surface.

This work demonstrates the scalability and applicability of autonomous directional drilling in unconventional UAE operations, setting a new benchmark for well construction performance while supporting national strategies for AI adoption and digital transformation.

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